

# Financial NLP and Sentiment Analysis

## Lecture 09 — Treating Language as Financial Data

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# Outline

- 1 Text as Financial Data
- 2 Sentiment Analysis Methods
- 3 Applications by Text Source
- 4 From Sentiment to Tradeable Signal
- 5 Practical Pipeline

# Why Treat Language as Data?

## THE BIGGER PICTURE

A single trading day now generates hundreds of thousands of news articles, social posts, filings, and analyst reports. Extracting the signal that moves prices — in real time, repeatably — is a defining problem of quantitative finance this decade.

Text is a data source with its own **measurement challenges**, exactly as:

- price data has bid–ask bounce,
- volume data has reporting lags,
- **text** has selection, duplication, and timestamp pitfalls.

## This lecture

Where financial text comes from → how to score its sentiment → what each source predicts → how to turn scores into a tradeable signal → the production pipeline.

# Five Sources of Financial Text

Sources differ on four axes: **frequency**, **curation**, **audience**, **regulatory oversight**.

| Source         | Curation / latency  | Canonical finding                  |
|----------------|---------------------|------------------------------------|
| Social media   | none / real-time    | Bollen 2011: Twitter mood → DJIA   |
| News & wires   | high / ms–hours     | Tetlock 2008: neg. words → returns |
| Earnings calls | scripted + Q&A      | Larcker 2012: tone → restatements  |
| SEC filings    | legal / quarterly   | Loughran–McDonald 2011             |
| Central banks  | extreme / scheduled | Schmeling 2019: tone → equities    |

## Source shapes method

The structural properties of each source determine both the information it carries *and* the right preprocessing / model. There is no one-size-fits-all pipeline.

# Source Properties That Matter

## Social media (X, Reddit/WSB, StockTwits)

- Least curated; high noise, sarcasm, bots
- Captures *retail attention* orthogonal to institutions
- Aggregates inform even when posts don't

## News & wires

- Reuters/Bloomberg: ms latency, structured event
- Key flaw: **selection** — only newsworthy firms

## Earnings calls

- Prepared remarks = soft-info framing
- Q&A reveals analyst concerns + candour

## SEC filings (EDGAR): 10-K (Item 1A, MD&A), 10-Q, 8-K, DEF 14A

- Disclosure law  $\Rightarrow$  truthful but vague

## Central banks: FOMC, ECB

- Meaning in calibrated *deviations* from boilerplate

## UNDER THE HOOD

Every pipeline normalises, deduplicates, and time-aligns before modelling. Skipping any of these biases the downstream sentiment measure.

## Tokenization & normalization

- Subword tokenizers: BPE (GPT) / WordPiece (BERT).
- Case folding for lexicons; *preserve* case for cased transformers.
- Number handling ([NUM]) — but EPS \$2.40 vs \$0.24 carries signal.
- Cashtag/ticker harmonisation (\$AAPL); boilerplate removal (XBRL, exhibits).

# Preprocessing: Deduplication & Temporal Alignment

**Deduplication** — most important, most neglected

- Wires reprint verbatim; retweets propagate identical text.
- Duplicates bias sentiment *and* inflate sample size → spurious narrow CIs.
- MinHash / LSH; mark dupes when trigram Jaccard  $> 0.6$ – $0.8$ .

**Temporal alignment** — avoid **look-ahead bias** (post-close news → same-day return) and **stale timestamps**; store publication *and* retrieval times separately.

# Three Coexisting Generations

## THE BIGGER PICTURE

No single method dominates. Lexicons are fast and interpretable; fine-tuned classifiers are strong in-domain; LLMs are flexible and need no labels. A competent practitioner uses all three and knows when each fits.

| Method               | Labels? | Strength           | Weakness            |
|----------------------|---------|--------------------|---------------------|
| Lexicon (LM)         | none    | transparent, cheap | no negation/context |
| Fine-tuned (FinBERT) | yes     | strong in-domain   | needs labelled data |
| LLM zero/few-shot    | none    | flexible, nuanced  | cost & latency      |



# Lexicon Method: Loughran–McDonald

## UNDER THE HOOD

Net sentiment ratio for document  $d$ :

$$S_{\text{LM}}(d) = \frac{N_{\text{pos}}(d) - N_{\text{neg}}(d)}{N_{\text{words}}(d)}$$

**Why a finance dictionary?** Harvard General Inquirer mislabels *tax*, *liability*, *cost*, *foreign*, *crude*, *cancer* as negative — neutral in 10-Ks. On 10-Ks, GI negativity tracks legal boilerplate, not performance.

**LM word lists** (built from the 10-K vocabulary):

| Negative | Litigious | Positive | Uncertainty | Weak modal | Strong modal |
|----------|-----------|----------|-------------|------------|--------------|
| 2,355    | 903       | 354      | 297         | 27         | 19           |

The **negative** list dominates — the asymmetry of disclosure language.

# LM Worked Example & Limits

## MD&A excerpt

*“Revenue declined 12%, reflecting challenging conditions and competitive pressure. The Company remains cautious but is confident in its long-term position.”*

- Negative hits: *declined, challenging, pressure, cautious* = 4
- Positive hits: *confident* = 1; total non-trivial words  $\approx 35$
- $S_{LM} = (1 - 4)/35 \approx -0.086$  (bottom-quartile MD&A)
- A GI classifier also flags *competitive*  $\rightarrow \approx -0.17$ : a distortion.

## Known limits

No negation (“not confident” scores positive); context-blind; domain shift (misses “headwinds”, “softness”). Still widely used: transparent, cheap, auditable.

# Fine-Tuned Classifiers: FinBERT & SEC-BERT

## UNDER THE HOOD

Take a pre-trained encoder, attach a head, fine-tune. For the [CLS] embedding  $\mathbf{h} \in \mathbb{R}^{d_h}$ :

$$P(y = c \mid d) = \text{softmax}(\mathbf{W}_c \mathbf{h} + \mathbf{b}_c), \quad c \in \{\text{pos, neu, neg}\}$$

minimising cross-entropy; predict  $\hat{y} = \arg \max_c P(y = c \mid d)$ .

## FinBERT (Araci 2019; Huang 2023)    SEC-BERT (Loukas 2022)

- Further-pretrain BERT on financial news + filings
  - Fine-tune on PhraseBank + analyst + call data
  - Beats GI, LM, and generic BERT
- Pre-trained *only* on EDGAR (10-K/Q/8-K)
  - Vocabulary tuned to legal-financial text
  - Wins in-domain on risk-factor / event tasks

**Edge over lexicons:**  $\mathbf{h}$  sees the whole sequence  $\Rightarrow$  handles negation

# The Financial PhraseBank

The standard labelled benchmark (Malo et al. 2014):

- **4,840** news sentences, labelled pos / neu / neg.
- **16** finance-trained annotators; majority label used.
- At 75% agreement:  $\approx 3,400$  high-confidence sentences, inter-annotator agreement  $\approx 75\text{--}80\%$ .
- **Neutral  $\approx$  half the corpus** — motivates macro-F1, not accuracy.

## Caveat

News-sentence style  $\neq$  filings / call language. Models tuned on PhraseBank may transfer poorly to long, syntactically complex documents — supplement with domain-matched labels.

# LLM Zero-Shot & Few-Shot Classification

## THE BIGGER PICTURE

GPT-4 / Claude / LLaMA-3 classify sentiment from a task description alone — *no fine-tuning, no labelled set*. Rapid prototyping on new domains.

**Zero-shot prompt sketch:** “*You are a financial analyst. Classify the sentiment . . . from the perspective of a stock investor.*” → “Neutral” with a rationale (revenue growth offset by margin compression).

## Benchmarks (real findings):

- Ko & Lee 2024: GPT-4 zero-shot *matches or exceeds* FinBERT on PhraseBank, FiQA, and a proprietary call dataset — with no labels.
- López-Lira & Tang 2023: GPT headline scores positively predict next-day returns, beating lexicons.
- Kirtac & Germano 2024: OPT (GPT-3 family) yields a markedly higher long-short Sharpe than the LM dictionary.
- Fatemi & Hu 2023: fine-tuned *small* LLMs hit SOTA; extra few-shot examples beyond a small set add no accuracy.

**Cost is the catch:** ~\$100–\$500 per million short sentences  $\Rightarrow$  distil into a small model for production.

# One Source, One Research Question

## THE BIGGER PICTURE

Each source has a natural question: does retail mood drive short-run prices? does news carry information beyond prices? does management tone reveal private information? does disclosure language predict performance? do policy words move asset prices?

| Source         | Question                                  |
|----------------|-------------------------------------------|
| Social media   | Retail attention → price pressure?        |
| News           | Incremental information beyond prices?    |
| Earnings calls | Tone → private information?               |
| SEC filings    | Disclosure language → future risk/return? |
| Central banks  | Linguistic cues → asset prices?           |

# Social Media: Attention, Disagreement, Bank Runs

## Evidence:

- Bollen 2011: “calm” Twitter mood Granger-causes DJIA (10-week sample; mixed replications).
- Cookson & Niessner 2020: StockTwits *disagreement* predicts volume and spreads.
- Gil-Bazo et al. 2025: 1.6M fund-family posts — tone *and* volume predict fund inflows.
- Cookson et al. 2026: SVB collapse — high Twitter-exposure banks lost more value; social media can *catalyse* bank runs (streaming use case).

**WSB / GameStop (Jan 2021):** coordinated WSB sentiment led GME/AMC by 1–2 days; mechanism via call demand → gamma-hedging amplification.

## Methodological hazards

Bots & coordinated inauthentic behaviour; severe selection; reverse causality (returns → posting).



## News (Tetlock 2007/08)

- WSJ pessimism → price pressure + reversion
- Firm-level neg. language → earnings
- Stale negativity adds no forecast power

## Earnings calls

- Larcker 2012: deception cues → restatements
- Price 2012: tone → post-call abnormal returns
- Cook & Kazinnik 2023: local LLMs ≈ closed-source, lower privacy risk

## SEC filings

- LM negativity ↓ filing-day return, ↑ volume
- Cohen 2020 “lazy prices”: unchanged 10-Ks underperform
- Lehner 2024: LLMs can rewrite filings to shift measured tone

## Central banks

- Schmeling 2019: positive tone → equities ↑
- Hansen 2018: deliberation quality → inflation outcomes
- Nakamura 2018: **information vs. policy** effect confound

# MD&A Readability: The Fog Index

## UNDER THE HOOD

$$\text{Fog} = 0.4 \left( \frac{N_{\text{words}}}{N_{\text{sentences}}} + 100 \cdot \frac{N_{\text{complex}}}{N_{\text{words}}} \right)$$

$N_{\text{complex}}$  = words with  $\geq 3$  syllables (excl. proper nouns, compound modifiers, simple verb inflections).

- $\text{Fog} = 12 \Leftrightarrow$  US high-school level.
- 10-K MD&A typically **18–22** — much harder than a newspaper.
- Li 2008: higher Fog  $\rightarrow$  lower current earnings, higher earnings volatility (managers obscure when performance is poor).
- Leheavy 2011: less readable 10-Ks  $\rightarrow$  fewer, less accurate analysts.

# Raw Scores Are Not Yet a Signal

## THE BIGGER PICTURE

A score per document is not tradeable. It needs aggregation, denoising, and statistical transformation. Two empirical frameworks test economic content: event studies and predictive regressions.

**Document** → **entity aggregation** (all docs on firm  $i$ , day  $t$ ):

- Simple mean:  $\bar{S}_t = \frac{1}{n_t} \sum_i S_i$
- Volume-weighted (reach / circulation)   ·   Recency-weighted (exp. decay)

Noisy sources (social) need weighting; curated sources (filings) do not.

## UNDER THE HOOD

Cross-sectional z-score within industry-year removes sector language fixed effects:

$$\tilde{S}_{i,t} = \frac{S_{i,t} - \mu_{j(i),t}}{\sigma_{j(i),t}}$$

( $j(i)$  = industry of firm  $i$ .) A defence firm and a software firm become comparable.

## Stationarity

- Long sentiment series drift (style change, universe change, regime shift).
- Apply ADF / KPSS; first-difference or detrend non-stationary series.

## Sentiment inflation

Re-scoring old text with a newer model changes measured sentiment. Use one *fixed* model vintage across all periods — or document the vintages.

## UNDER THE HOOD

Abnormal return  $AR_{i,\tau} = R_{i,\tau} - \hat{R}_{i,\tau}$  (benchmark, e.g. CAPM). Cumulative over  $[a, b]$ :

$$CAR_i[a, b] = \sum_{\tau=a}^b AR_{i,\tau}, \quad \overline{CAR}[a, b] = \frac{1}{N} \sum_i CAR_i[a, b].$$

**Does text beat the EPS number?** Condition on the earnings surprise:

$$CAR_i[1, 60] = \alpha + \beta_1 SUE_{i,0} + \beta_2 \tilde{S}_{i,0} + \varepsilon_{i,0}$$

with  $SUE = (EPS - \widehat{EPS})/\hat{\sigma}$ . Incremental text content  $\Rightarrow \hat{\beta}_2 > 0$ .

**High-frequency FOMC:** 30-min window around announcements (Nakamura 2018) demands sentence-level scoring *inside* the window  $\rightarrow$  low-latency inference.

# Two Horizons, Two Mechanisms

|            | Short horizon                             | Long horizon                 |
|------------|-------------------------------------------|------------------------------|
| Channel    | attention / price pressure                | fundamental information      |
| Evidence   | Da 2015: attention $\rightarrow$ reversal | Tetlock 2008; Cohen 2020     |
| Data needs | real-time, low-latency                    | daily/weekly, heavier models |
| Audience   | HFT-adjacent industry                     | most academic research       |
| Alpha      | larger (pre-cost)                         | smaller, more accessible     |

## Practical implication

Short-horizon strategies need real-time feeds and execution within minutes; long-horizon strategies tolerate batch processing and richer models. Industry chases the short horizon (more alpha); academia studies the long horizon (lower data/execution cost).

# From “What” to “How”

## THE BIGGER PICTURE

Earlier sections said what to extract and why. The pipeline is the engineering: collect text at scale, run inference in production, evaluate classifier quality with metrics that don't mislead.

### Three production concerns:

- ➊ **Data collection** — EDGAR, social APIs, news vendors.
- ➋ **Inference at scale** — batch (throughput) vs. streaming (latency).
- ➌ **Evaluation** — label quality *and* classifier quality *and* economic validity.

(Collection & scale engineering detailed in the appendix.)

# Evaluation: Annotator Agreement

## UNDER THE HOOD

Krippendorff's alpha measures agreement beyond chance:

$$\alpha = 1 - \frac{D_o}{D_e}$$

$D_o$  = observed,  $D_e$  = expected disagreement. Nominal  $k$ -class:  $D_e = 1 - \sum_c (n_c/n)^2$ .  $\alpha = 0$ : chance;  $\alpha = 1$ : perfect.

Thresholds:  $\alpha \geq 0.667$  reliable;  $\geq 0.800$  high confidence. Financial benchmarks: **0.65–0.78** — genuine ambiguity of finance sentences.

| Dataset                                | Classes | Alpha          |
|----------------------------------------|---------|----------------|
| Financial PhraseBank (100% agr.)       | 3       | 1.00           |
| Financial PhraseBank (66% agr.)        | 3       | $\approx 0.72$ |
| Earnings call sentences (practitioner) | 3       | 0.68–0.75      |
| FOMC statement sentences               | 3       | 0.70–0.78      |



# Evaluation: Classifier Metrics & Economic Validity

## UNDER THE HOOD

Per class  $c$ :

$$P_c = \frac{TP_c}{TP_c + FP_c}, R_c = \frac{TP_c}{TP_c + FN_c}, F1_c = \frac{2P_cR_c}{P_c + R_c}.$$

Report **macro-F1** (unweighted mean over classes).

## Why macro, not accuracy

Neutral  $\approx$  half of financial sentences. An accuracy-maximiser that over-predicts neutral scores high while failing on the economically important positive / negative classes.

**Economic validity  $\neq$  statistical metrics.** Macro-F1 = 0.75 can still have zero alpha if the test set is unrepresentative or labels are misaligned. Best practice: regress aggregate sentiment on forward returns / earnings

# Lecture 09 Summary

- **Sources:** social → filings, spanning curation and oversight; structure dictates preprocessing and model.
- **Methods:** lexicon (LM, the filings standard), fine-tuned (FinBERT/SEC-BERT), LLM zero/few-shot — LLMs match or beat fine-tuned without labels, at higher cost.
- **Applications:** attention (social), information (news), soft disclosure (calls), risk (filings), policy signal (central banks).
- **Signal:** aggregate → z-score within industry-year → test stationarity; event study + predictive regression assess economic content.
- **Pipeline:** batch + streaming inference; evaluate with Krippendorff's  $\alpha$ , macro-F1, *and* a held-out economic test.

## Next

Practical 09 — LM scoring, FinBERT vs. lexicon, and a filing-date event study on real EDGAR + price data.

# APPENDIX

## Data Collection, Inference at Scale, and Extensions

Extra / optional material — beyond the core lecture.

## **EDGAR full-text search** (`efits.sec.gov`)

- 1 Query form types (10-K, 8-K) and date range.
- 2 Download index pages → document links.
- 3 Parse primary HTML/SGML (BeautifulSoup, `sec-edgar-downloader`).
- 4 Extract MD&A / Risk Factors by header matching.
- 5 Store Parquet, indexed by CIK / period / filing date.

**Twitter/X Academic API** (paid since 2022): filters `has:cashtags`, `lang:en`, `verified-author`. Alternatives: StockTwits API, Reddit Pushshift (restricted), commercial sentiment feeds.

**News vendors:** Bloomberg, Dow Jones DNA, Refinitiv, RavenPack (machine-readable event records); GDELT (free, noisier, 15-min updates) for research.

## **Batch** (throughput-bound):

- FinBERT (110M), 512 tokens, single A100: hundreds of docs/sec; multi-million archive in hours.
- Optimisations: dynamic padding/batching; FP16/BF16; distillation (110M  $\rightarrow$  33M,  $< 5\%$  accuracy loss); data/CPU parallelism.

## **Streaming** (latency-bound):

- Target  $< 100$  ms for FOMC; near-real-time for live call transcription.
- Kafka/Kinesis  $\rightarrow$  ONNX inference workers  $\rightarrow$  downstream aggregation.
- `onnxruntime` + quantised weights:  $< 100$  ms/sentence on modern CPUs.

## Appendix: Ordinal Krippendorff's Alpha

For ordered labels (pos > neu > neg) the nominal coincidence metric is replaced by a weighted disagreement:

$$d^2(c, c') = \left( \sum_{g=c}^{c'} n_g \right)^2,$$

penalising disagreements by the number of scale steps between labels ( $n_g$  = frequency of category  $g$ ). Ordinal  $\alpha$  is more conservative than nominal  $\alpha$  for sentiment corpora — a stricter lower bound on agreement.

**News-Implied Volatility (Manela & Moreira 2017):** project WSJ front-page text onto realized VIX variation; the measure extends to **1890**, capturing disaster-risk eras (Depression, WWII, 2008–09) far beyond VIX's 1990 start.

**News in factor models (Ke et al. 2019):** supervised extraction of price-relevant news features adds predictive content, esp. at short horizons; intraday alpha captured by HFT, longer-horizon signals lower-Sharpe but accessible.

**Climate / ESG (Engle et al. 2020):** a climate-news index builds a hedge portfolio; high climate-news exposure → lower returns on high-news days. Generalises to pandemic / geopolitical / supply-chain risk indices.

## Beyond the context window:

- Chiu & Hung 2024: finance-specific LLaMA-2 + LLM summarisation of long 10-Ks → higher buy-and-hold abnormal returns than FinBERT.
- Siano 2025: LLM word-context embeddings of earnings press releases explain far more return variation than dictionaries; raise  $R^2$  of financial-statement surprises.
- Xu & Babaian 2025: FinBERT + GPT-4o *ensemble* on FOMC press conferences — cross-family combination is more stable than any single model.